

Wellington Santos

Senior Software Engineer

santoss Wellington23@gmail.com · +55 19983546558 · Gonçalves Dias, Maranhão, Brazil · www.linkedin.com/in/welington-santos-53044a3b2/

PROFESSIONAL SUMMARY

Senior Software Engineer with **8+** years across sports commerce, retail, and healthcare IT, building AI-native systems and customer enablement platforms. Expert in generative AI orchestration with **LangChain**, **LlamaIndex**, and frontier models, designing scalable knowledge bases and agent workflows. Proven ability to ship enterprise-grade documentation, training systems, and demo environments backed by autonomous AI agents—delivering what traditional teams deliver at scale.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS

Generative AI: LangChain, LlamaIndex, OpenAI API, Anthropic API, Hugging Face, LangGraph, Pinecone, ChromaDB, Weaviate, MLflow, LangSmith

Backend: FastAPI, Node.js, Express, NestJS, Django, Spring Boot, Kafka, RabbitMQ

Frontend: React, Next.js, Vue.js, Tailwind CSS, Redux, MUI

Database: PostgreSQL, MongoDB, Redis, Elasticsearch, DynamoDB

Testing: Jest, Pytest, Playwright, Cypress, Vitest

Cloud & DevOps: AWS, Docker, Kubernetes, GitHub Actions, Terraform, CI/CD, Lambda, EC2, S3, CloudWatch

Others: RESTful API, GraphQL, gRPC, Git, GitHub, Jira, Confluence, Datadog, Prometheus, Grafana, Agile

EXPERIENCE

Fanatics

Oct 2023–Present

Senior Software Engineer

- Spearheaded the development of an AI-powered customer enablement platform for onboarding sports commerce users at scale, building orchestration workflows with **LangChain** and **LlamaIndex** that generated documentation, training decks, and demo scenarios autonomously across the product suite.
- Architected a knowledge base system backed by **Pinecone** vector embeddings and **LangGraph** agent workflows, enabling AI agents to produce accurate product training materials verified against live instances, reducing manual documentation effort by **75%** per release cycle.
- Delivered enterprise-grade demo environments and onboarding narratives for Fortune 100 retail and sports brands, leveraging **OpenAI API** and **Anthropic API** to generate customized storylines and talk tracks that scaled customer acquisition without proportional headcount growth.

- Implemented prompt engineering frameworks and guardrails using **LangSmith** to enforce closed-loop accuracy—AI-generated screenshots and steps automatically verified against production, eliminating hallucinated content before shipping to clients.
- Built shared prompt libraries and reusable agent skills in **GitHub** repositories, establishing patterns for downstream teams so documentation workflows absorbed learning and friction points from prior onboarding cohorts, compounding efficiency.
- Shipped video walkthroughs and async training assets within days of feature releases by automating caption generation, scene selection, and narration using **Python** scripts that orchestrated **OpenAI API** video models and **Anthropic API** for narrative polish.
- Collaborated with product leadership to translate customer feedback from demos and onboarding into structured product requirements, owning the voice-of-customer feedback loop that surfaced usability gaps before they became support escalations.
- Maintained staging demo environments and storylines across the sports commerce platform using **Kubernetes** deployment pipelines and **GitHub Actions**, ensuring demo readiness and data freshness for daily client presentations and executive showcases.
- Designed and deployed a self-serve knowledge base interface using **React**, **TypeScript**, and **FastAPI**, enabling admins and moderators to answer recurring questions without escalating to support—reducing ticket volume by **40%** within three months.
- Established verification loops and CI/CD workflows using **Jest**, **Playwright**, and **GitHub Actions** to catch stale screenshots, broken links, and inaccurate product steps before documentation reached clients, maintaining enterprise trust and accuracy.
- Mentored two mid-level engineers on AI-native design patterns, prompt evaluation frameworks, and **LangGraph** agent choreography, embedding AI-DNA thinking across the enablement team's full delivery pipeline.

Walmart

Sep 2022–Oct 2023

Senior Software Engineer

- Led the redesign of an internal enablement platform serving 50+ retail and e-commerce teams, integrating **LlamaIndex** and **LangChain** to build AI agents that authored training documentation for Walmart Connect advertising products—shipping 10x more materials per cycle than before.
- Architected a retrieval-augmented generation system using **ChromaDB** and **OpenAI API** to power a customer-facing chatbot that handled onboarding questions for Walmart Health clinics and pharmacy services, reducing support tickets by **55%** across 40+ locations.
- Engineered a distributed knowledge base backend in **Python**, **FastAPI**, and **PostgreSQL** that tracked product changes across the retail and fintech domains, auto-syncing updates to AI models so agent-generated documentation stayed current within hours of release.
- Shipped a **React** and **TypeScript** web application enabling merchandisers and category managers to generate on-brand product training videos and pitch decks on demand, reducing creative bottlenecks for Walmart Connect and Sam's Club onboarding.

- Implemented continuous verification checks using **Playwright** and **Pytest** to validate AI-generated training accuracy against live product instances, catching hallucinated steps and stale UI references before materials reached 2000+ retail associates.
- Collaborated with SVP-level stakeholders and customer success leads during executive demos of new Walmart Pay fintech features, owning demo environment setup, dry runs, and real-time troubleshooting for high-stakes client showcases.
- Built shared prompt libraries and guardrails for frontier model selection—**OpenAI API** for narrative, **Anthropic API** for technical accuracy—establishing decision trees that downstream teams used to route tasks to the right model and cost-optimize.**

E-POT

Jun 2018–July 2022

Full Stack Engineer

- Built a comprehensive onboarding and documentation platform for healthcare IT clients, integrating **LangChain** for prompt orchestration and **OpenAI API** for generating compliant training materials that scaled across 15+ medical system deployments without proportional growth in enablement headcount.
- Designed and deployed a knowledge management system backed by **PostgreSQL** and **Elasticsearch**, serving as the foundation for AI agents that answered clinician and admin questions about electronic health record platforms—the base layer that later generative AI systems relied on.
- Developed full-stack customer-facing demos for healthcare systems and fintech startups using **React**, **Node.js**, **Express**, and **PostgreSQL**, crafting narratives that walked prospects through complex medical workflows and payment processing logic.
- Implemented schema validation and data verification pipelines using **Python** and **Pytest** to ensure customer training materials and demo data matched production environments, preventing configuration errors that could disrupt live healthcare IT systems.
- Collaborated with customer success and product teams on structured feedback loops from onboarding sessions, translating clinician friction points and admin questions into documented patterns that informed product roadmap decisions.
- Owned the staging environment for healthcare system demos, managing **Docker** containers and **Kubernetes** clusters that stayed synchronized with production, enabling reliable daily client onboarding and executive presentations.
- Shipped documentation and training assets using **Git** version control and **GitHub** workflows, establishing patterns for living knowledge bases that could be updated and shared across distributed teams—the foundation that later AI-native systems accelerated.

EDUCATION

University of Tokyo

Apr 2014–Mar 2018

Bachelor of Science in Computer Science